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Coulomb dissociation of ^8B and ^6Li

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Abstract

The analysis of high-energy Coulomb dissociation experiments with ^8B and ^6Li beams in order to extract low-energy S factors of the $^7\text{Be}(p,\gamma)^8\text{B}$ and $^4\text{He}(d,\gamma)^6\text{Li}$ radiative capture reactions relies on a precise knowledge of the reaction mechanism. A comparison of various experimental cross sections with predictions from theory allows us to identify the relevant ingredients that are necessary for a successful theoretical description. The cases of ^8B and ^6Li Coulomb breakup show substantial differences related to the structure of the nuclei.

1. Introduction

Since the proposal of the Coulomb dissociation method as an independent approach to determine radiative cross sections of astrophysical interest at low energies more than 20 years ago [1], the field has experienced a steady growth with many experimental and theoretical studies focusing on various nuclei and on different aspects of the reaction mechanism. For a reliable application of the method to cases where there are no direct experiments feasible and for a proper understanding of the reaction, a study of selected test cases with a comparison to available direct measurements is crucial. It is important to assess the validity of the theoretical description and to identify the breakup mechanisms that are relevant in the analysis of the Coulomb dissociation experiments for each case.

In this contribution, the Coulomb dissociation of ^8B and ^6Li at high projectile energies is compared from a theoretical point of view. These two systems exhibit very different properties that have a big impact on the applicability of the indirect approach to extract low-energy S factors. Combining the information from various observables, i.e., the energy and angular dependence of the cross sections, it is possible to estimate whether the conditions for a successful application of the Coulomb dissociation method are met. The experience gained in the breakup studies of ^8B is very valuable for the analysis of ^6Li dissociation experiments.

2. General aspects and theory of the Coulomb dissociation method

The main idea of the Coulomb dissociation method is to replace a radiative capture reaction $c(b, \gamma)a$ by its inverse photo absorption reaction $a(\gamma, b)c$ and to substitute the real photons in the latter process by virtual photons during the scattering of a projectile a on a heavy target nucleus X , i.e., studying the breakup reaction $X(a, bc)X$. The cross sections of the first two reactions are related by the theorem of detailed balance and the breakup cross section can be expressed by the photodissociation cross section and virtual photon numbers if a single-photon exchange in the breakup is assumed. The equivalent photon numbers are supplied by theory. Therefore, the application of the indirect method and the relation of cross sections seems to be straightforward in principle. The large flux of virtual photons and kinematical transformation factors yield much larger absolute values for cross sections of the breakup reaction as compared to the capture reaction. High projectile energies allow us to reach high excitation energies in the projectile system. The contribution of different electromagnetic multipoles depends on the kinematical conditions generally favoring E2 over E1 and suppressing M1 transitions.

In the actual application of the indirect method, however, there are a number of factors that complicate the analysis of experimental breakup data, and the extraction of radiative capture cross sections becomes more involved. In practice, the theoretical description of the breakup reaction starts with establishing a structure model for the bound and scattering states of the projectile system $a = b + c$. It is adjusted to give reasonable values for various system properties, e.g., binding and resonance energies, scattering lengths, and for the radiative capture cross section of interest. This model is combined with a breakup reaction model in order to calculate dissociation cross sections that can be compared with experimental data. In the reaction model, different contributions to the breakup process can be considered depending on their relevance for the individual case. It can be necessary to include higher-order processes (multi-photon exchange or ‘post-acceleration’) in the electromagnetic excitation or nuclear breakup contributions. In a next step, the theoretical cross sections from the reaction model have to be corrected for the experimental conditions, e.g., detector efficiency and acceptance, finite resolution, etc. Finally, from the deviation between theoretical and experimental cross sections for the breakup reaction, modifications of the structure model and of the radiative capture cross section of interest have to be inferred.

In order to facilitate the analysis process, a single computer program CDXS+ has been developed for medium to high-energy breakup reactions that contains all the relevant steps in the theoretical description with many additional features. See the manual for details [2]. The projectile system is described in a two-body cluster model with a choice of different interaction potentials for bound and continuum states. For the electromagnetic contribution to the breakup, E1, E2 and M1 transitions can be considered in nonrelativistic or relativistic semiclassical approximations with possible quantal diffraction corrections. Nuclear breakup and absorption are treated in a prior-form distorted-wave Born approximation (DWBA) with eikonal wavefunctions where optical potentials from systematics or from folding approaches can be selected. A large variety of single, double and triple differential breakup cross sections are calculated. Furthermore, it is possible to generate event distributions in a Monte Carlo simulation from the most general breakup cross section. This allows us to apply corrections for experimental conditions in a simple way. For example, the event distributions can be used as an input to GEANT simulations of the experimental setup. In the following two applications of this computer code are presented.

3. ${}^7\text{Be}(p,\gamma){}^8\text{B}$ and Coulomb dissociation of ${}^8\text{B}$

The radiative capture reaction ${}^7\text{Be}(p,\gamma){}^8\text{B}$ appears in the third branch of the pp-chain. A precise value for the capture cross section at the relevant energies of about 20 keV is essential to predict the flux of high-energy neutrinos from the subsequent β -decay of ${}^8\text{B}$ that is observed on earth [3, 4]. There have been a large number of direct experiments [5–15] in the past, where the earlier ones showed a large scattering of the extracted zero-energy S factor. It was thought that the study of ${}^8\text{B}$ breakup could give important information from an independent approach with different systematic errors. Consequently, a large number of Coulomb dissociation experiments [16–21] were performed at various projectile energies. However, the focus of these indirect measurements has shifted in recent years because the latest direct experiments with very precise data allow us to check the validity of the indirect method in the case of ${}^8\text{B}$ breakup.

For a detailed comparison of theoretical predictions with measured data from an indirect experiment, it is crucial that highly-differential cross sections are available depending on excitation energy and various angles of the outgoing fragments. These conditions are ideally fulfilled for the second GSI experiment [20, 21] with a 254 A MeV secondary beam of radioactive ${}^8\text{B}$ on a lead target with complete kinematics and high statistics. Unfortunately, in the first GSI experiment [18] it was not possible to extract these cross sections with sufficient precision. Details of the second GSI experiment and of the theoretical description can be found in [21]. Here, it suffices to summarize the main features of the calculation and the results of the analysis.

The ground state of ${}^8\text{B}$ with total angular momentum and parity $J^\pi = 2^+$ and a proton separation energy of only 137 keV can be considered as a halo state with a proton–core relative wavefunction that extends to large distances. It is well described in a ${}^7\text{Be}$ -p cluster model with a p-wave for the relative motion. Radiative capture from the ${}^7\text{Be}$ -p continuum at low energies to this bound state is dominated by non-resonant E1 transitions from s-waves and to a lesser extent from d-waves. E2 transitions from p- and f-waves have a negligible contribution at these energies of astrophysical interest since they are more than three orders of magnitude smaller, see. e.g., figure 2 in [21]. The strong resonant M1 transition of the $J^\pi = 1^+$ state at 632 keV c.m. energy in the continuum is also not relevant for astrophysics but appears clearly in the directly measured capture cross sections. With a careful adjustment of the potential parameters for the bound and scattering states in the two-body cluster model, the non-resonant part of the astrophysical S factor of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ reaction is well described in comparison with direct data and more microscopic theoretical models. The resonant M1 contribution is not well reproduced in the simple two-body model but with a modification of the parameters in this particular channel and a rescaling of the M1 amplitude a reasonable agreement with the observation can be achieved.

For the analysis of the experiment, Coulomb dissociation cross sections were calculated in an eikonal-corrected first-order relativistic semiclassical approximation taking into account the absorption of the projectile on the target. They were converted to event distributions that served as an input to GEANT simulations. A comparison of the theoretical predictions with the experimental data leads to the following conclusions. The resonant M1 contribution to the breakup is strongly suppressed but it is observed in the excitation function for the conditions of a 254 A MeV ${}^8\text{B}$ beam in contrast to the other Coulomb dissociation experiments at lower beam energies. The dependence of the cross section on the ${}^8\text{B}$ -Pb scattering angle can only be described if the absorption of ${}^8\text{B}$ on the target is taken into account. Then the shape of the angular distribution is well reproduced if only E1 transitions from the ground state are considered. There is no sign of a substantial E2 contribution in this particular cross section.

The ${}^7\text{Be}$ -p relative angle distribution from theory is close to the experimental data. Thus the structure model seems to need only small readjustments for the S -factor justifying its use in the simulation. From a comparison of the fragment angular distributions in the ${}^8\text{B}$ c.m. system there is also no clear sign that E2 contributions or higher-order effects, that are anyhow expected to be small at high projectile energies, are needed in the theory to describe the experiment. The structure model with only E1 excitations seems to be sufficient and an E2 contribution, even if amplified in the breakup process, is not required. Possible contributions to the breakup from nuclear excitation have not been considered so far. It remains a task for the future to study their effect but nuclear breakup is expected to contribute with a similar strength like the E2 excitation. The energy dependence of the extracted S factor follows the direct data and the rescaled results of the microscopic cluster model of [22]. See figures 15 and 16 in [21]. The zero-energy S factor $S_{17}(0) = 20.6 \pm 0.8(\text{stat}) \pm 1.2(\text{syst}) \text{ eV b}$ extracted from the experimental data by an extrapolation with the energy dependence of the same cluster model is consistent with the most precise direct results. In summary, the Coulomb dissociation method has proven to work in the particular case of ${}^8\text{B}$, a weakly-bound nucleus where E1 transitions to the continuum dominate the cross sections.

4. ${}^4\text{He}(d,\gamma){}^6\text{Li}$ and Coulomb dissociation of ${}^6\text{Li}$

The element lithium is observed with a very small abundance in the universe. It is assumed to be produced in Big Bang nucleosynthesis (BBN). From the observation of fluctuations of the cosmic microwave background by WMAP [23] all parameters of standard Big Bang nucleosynthesis are fixed and definite predictions for primordial element abundances can be made, see, e.g., [24]. There is an overall agreement between observation and theory for the light nuclei ${}^2\text{H}$, ${}^3\text{He}$ and ${}^4\text{He}$ but the predicted abundances for the two stable Lithium isotopes disagree with the observation constituting the BBN lithium problem.

The amount of primordial ${}^6\text{Li}$ is determined by the radiative capture reaction ${}^4\text{He}(d,\gamma){}^6\text{Li}$ with a large uncertainty at low energies. Direct measurements at the 3^+ resonance near 711 keV for the α -d c.m. energy [25] and at energies in the non-resonant continuum above [26] are consistent with potential model calculations. But the results for the low-energy S factor extracted from a Coulomb dissociation experiment [27] with a 26 A MeV ${}^6\text{Li}$ beam deviate strongly from the theoretical energy dependence at very low energies. Incidentally, ${}^6\text{Li}$ is one of the first nuclei where it was tried to apply the indirect Coulomb breakup method (at 10 A MeV projectile energy) in order to extract a radiative capture cross section [28].

The capture cross section of the reaction ${}^4\text{He}(d,\gamma){}^6\text{Li}$ is dominated by E2 transitions from d-waves in the continuum to the 1^+ s-wave ground state of ${}^6\text{Li}$ that is well described as an α -d two-body cluster system. Because of the different penetrabilities of s- and d-waves, the S factor for capture to the ${}^6\text{Li}$ ground state shows a much steeper energy dependence as compared to the S factor for the capture to ${}^8\text{B}$ that is almost constant at low energies. A possible E1 contribution to the ${}^4\text{He}(d,\gamma){}^6\text{Li}$ cross section is isospin forbidden (α and d have almost the same charge-to-mass ratio) and can compete with E2 only at extremely low energies due to the different energy dependence of the cross sections. Adjusting the α -d potentials in the potential model for the bound state to the binding energy of 1.474 MeV and for the continuum states to the resonance energy, an E2 S factor is obtained that nicely follows the direct data, see figure 1. In contrast, the disagreement with the indirect data at very low energies is obvious. A new breakup experiment with a ${}^6\text{Li}$ beam of higher energy could help us to clarify the situation. The data of a recent Coulomb dissociation experiment at GSI with 150 A MeV ${}^6\text{Li}$ projectiles on a lead target [29] are currently analyzed. Here, the main difficulties in the theoretical description are discussed.

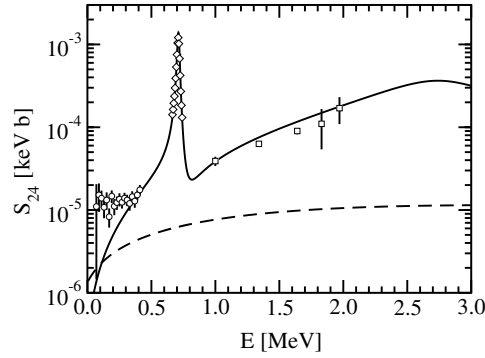


Figure 1. Astrophysical S factor of the ${}^4\text{He}(d,\gamma){}^6\text{Li}$ capture reaction in the potential model (solid line: E2, dashed line E1) in comparison with direct data (diamonds [25], squares [26]) and indirect data (circles) from the Coulomb breakup experiment [27].

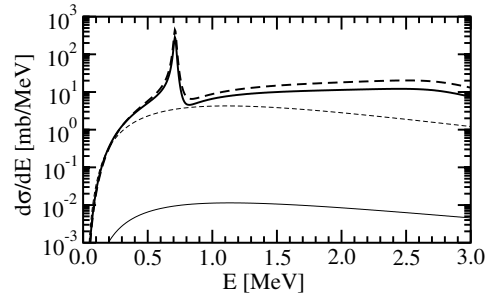


Figure 2. Dependence of the cross section on the α -d c.m. energy for the breakup of ${}^6\text{Li}$ into α and d for a 150 A MeV ${}^6\text{Li}$ beam on a Pb target. Electric transitions are denoted by solid and nuclear transitions by dashed lines. Thin lines correspond to dipole and thick lines to quadrupole excitations.

There are several important differences when the conditions for the ${}^6\text{Li}$ and ${}^8\text{B}$ breakup are compared. The relative enhancement of E2 transitions will make it almost impossible to detect a possible E1 contribution in the ${}^6\text{Li}$ dissociation whereas for ${}^8\text{B}$ E1 was dominating and E2 could be increased to a sizable fraction of the breakup cross section. Higher-order effects are expected to be small for ${}^6\text{Li}$ breakup because of the equal charge-to-mass ratio of ${}^4\text{He}$ and ${}^2\text{H}$. The higher binding energy of ${}^6\text{Li}$ and the lower beam energy in the experiment reduces the probability for an electromagnetic breakup and nuclear contributions can become important. Therefore, nuclear breakup and absorption was included in the calculation of the cross sections with the computer code CDXS+ [2] employing the eikonal DWBA. The relevant optical potentials for the projectile-target and fragment-target interaction were derived from a folding of the projectile and fragment densities with relativistic nucleon-target potentials [30]. They allow a reasonable description of elastic scattering of ${}^4\text{He}$ and ${}^2\text{H}$ on a lead nucleus at lower energies where experimental data are available. However, for ${}^6\text{Li}$ a significant rescaling of the real potential (≈ 0.5) is required. There is definitely a need for elastic scattering cross sections at the relevant higher beam energies to have a better constraint of the optical potential parameters.

In figure 2 the individual contributions to the breakup cross section from electric and nuclear dipole and quadrupole excitations are shown as a function of the α -d c.m. energy,

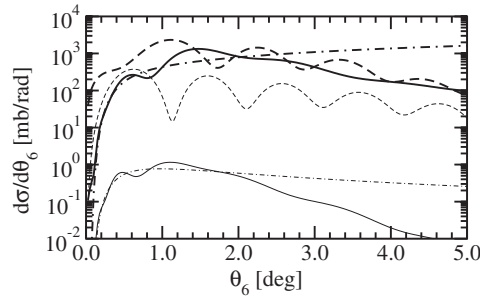


Figure 3. Dependence of the cross section on the ${}^6\text{Li}$ -Pb scattering angle for the breakup of ${}^6\text{Li}$ into α and d for a 150 A MeV ${}^6\text{Li}$ beam on a Pb target integrated from 0 to 3 MeV c.m. energy. Electric transitions are denoted by solid and nuclear transitions by dashed lines. Thin lines correspond to dipole and thick lines to quadrupole excitations. Results of a pure semiclassical Coulomb breakup calculation without eikonal corrections are shown with dash-dotted lines.

integrated over the projectile–target scattering angles from zero to about 5° . As expected, the E1 contribution is strongly suppressed with respect to E2 when the ratio of dissociation cross sections is compared to the case of radiative capture in figure 1. Nuclear contributions to the breakup are very large in the present calculation with a similar magnitude as the E2 excitation. There is no suppression of the nuclear dipole excitation since it is not forbidden by selection rules.

The dependence of the breakup cross section on the ${}^6\text{Li}$ -Pb scattering angle is displayed in figure 3 for the individual contributions as in figure 2. Here the range for the integration in c.m. energy was chosen from 0 to 3 MeV. A similar picture as from the previous figure emerges: the suppression of the E1 contribution and the strong nuclear breakup components that compete with the E2 excitation. In addition, diffraction effects due to the absorption on the target show up with clear oscillations of the cross section. At larger scattering angles, the absorption also leads to a reduction of the breakup cross section. These two effects are absent in the pure semiclassical description of the electromagnetic excitation without the quantal eikonal corrections.

With the strong nuclear component to the breakup, it seems very difficult to extract a reliable S factor for the radiative capture reaction ${}^4\text{He}(d,\gamma){}^6\text{Li}$ from the Coulomb dissociation experiment. However, it has to be checked whether the nuclear breakup component as predicted in the current theoretical model is realistic. On the one hand, the employed optical potentials should be tested with elastic scattering cross section measured at the actual energy of the experiment. On the other hand, it is necessary to compare various measured breakup cross sections with the theoretical predictions in order to identify observables that are most sensitive to the nuclear contribution and that allow us to constrain the nuclear breakup contribution. To this end the theoretical cross sections were converted to event distributions for various observables that allow an easy comparison with the experiment. (Experimental corrections like finite resolution, efficiency etc are not yet included.) As seen in figure 4, the c.m. energy distribution or the closely related distribution of the α -d relative angle cannot distinguish between electromagnetic and nuclear breakup because both have an almost identical shape. There is a chance, however, for the ${}^6\text{Li}$ angular distribution where the Coulomb and nuclear breakup exhibit different shapes. A comparison of this theoretical prediction with actual experimental data can help us to validate the present model calculations or to hint at modifications that are required. Maybe it is necessary to look also into specific bins of c.m.

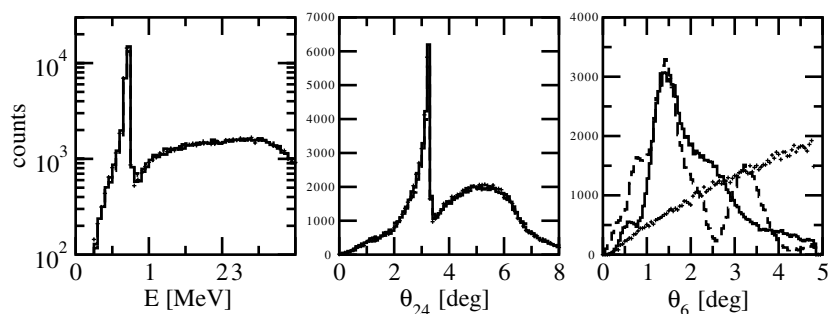


Figure 4. Distribution of various observables generated from the theoretical model for the breakup of ${}^6\text{Li}$ into α and d for a 150 A MeV ${}^6\text{Li}$ beam on a Pb target. Left: α -d c.m. energy distribution, center: α -d relative angle distribution, right: ${}^6\text{Li}$ scattering angle distribution. The eikonal corrected Coulomb breakup calculation is denoted by dashed lines and Coulomb+nuclear breakup by solid lines. The crosses correspond to a pure semiclassical Coulomb breakup calculation without eikonal corrections.

energy and scattering angles in order to obtain more detailed information. The present analysis of the ${}^6\text{Li}$ breakup experiment will hopefully give an answer to the question of the importance of the nuclear breakup component.

5. Conclusions

The ${}^8\text{B}$ breakup experiment at GSI with a 254 A MeV projectile beam [20, 21] showed that it is possible to extract low-energy capture cross sections of astrophysical interest with the indirect Coulomb dissociation approach that are consistent with direct experiments. In this case the experimental conditions and the properties of the nuclear system were favorable. In comparison, the breakup of ${}^6\text{Li}$ at 150 A MeV [29] seems to be a challenging case for extracting information on the ${}^4\text{He}(d,\gamma){}^6\text{Li}$ capture cross section because the nuclear breakup component is estimated to be strong in the present theoretical calculation. However, this result has to be checked carefully and tested against experimental data before a final conclusion can be drawn.

With the experience of these two high-energy breakup experiments, it becomes obvious that theoretical support is essential for a reliable application of the indirect Coulomb dissociation method. For the preparation of future measurements, theoretical calculations can give valuable hints for the selection of suitable reactions and proper experimental conditions.

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